API Endpoint Plan

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/auth/register	Registers a new user account. Password is hashed before storage.	Public	FirstName, LastName, Email, Password	201 Created. Returns UserID, name, email, role and CreatedAt. 400 Bad Request – missing or invalid fields. 409 Conflict – email already registered.
POST	/api/auth/login	Authenticates a user and returns an authentication token.	Public	Email, Password	200 OK. Returns access token, UserID, name, email and role. 400 Bad Request – missing fields. 401 Unauthorized – invalid credentials.
GET	/api/users/me	Returns the profile of the currently authenticated user.	Authenticated	None	200 OK. Returns UserID, FirstName, LastName, Email, Role, CreatedAt. 401 Unauthorized – no valid token provided.
PUT	/api/users/me	Updates the currently authenticated user's profile information.	Authenticated	FirstName, LastName, Email	200 OK. Returns the updated user profile. 400 Bad Request – validation errors. 401 Unauthorized – no valid token provided. 409 Conflict – email already in use.
PUT	/api/users/me/password	Changes the authenticated user's password.	Authenticated	currentPassword, newPassword	200 OK. Returns password update confirmation. 400 Bad Request – new password fails validation rules. 401 Unauthorized – currentPassword is incorrect, or no valid token.

GET	/api/users/{userId}	Retrieves a specific user's profile. Useful for viewing organiser information.	Admin	None	200 OK. Returns UserID, FirstName, LastName, Email, Role and CreatedAt. 401 Unauthorized – no valid token. 403 Forbidden – caller is not an Admin. 404 Not Found – user does not exist.
GET	/api/users	Retrieves a list of registered users.	Admin	None	200 OK. Returns a list of users. 401 Unauthorized – no valid token. 403 Forbidden – caller is not an Admin.
GET	/api/events	Retrieves all available events.	Public	None	200 OK. Returns a list of events with event name, description, date, venue, organiser and creation date.
GET	/api/events/{eventId}	Retrieves detailed information about one event.	Public	None	200 OK. Returns event details, venue, organiser and categories. 404 Not Found – event does not exist.
POST	/api/events	Creates a new event. The authenticated user becomes the organiser.	Admin, Organiser	VenueID, Name, Description, EventDate	201 Created. Returns the newly created EventID and event details. 400 Bad Request – validation errors. 401 Unauthorized – no valid token. 403 Forbidden – caller is not Admin/Organiser. 404 Not Found – VenueID does not exist.
PUT	/api/events/{eventId}	Updates an existing event.	Admin, Organiser	VenueID, Name, Description, EventDate	200 OK. Returns the updated event. 400 Bad Request – validation errors. 401 Unauthorized – no valid token. 403 Forbidden – caller is not Admin or the event's Organiser. 404 Not Found – event does not exist.

DELETE	/api/events/{eventId}	Deletes an event.	Admin, Organiser	None	204 No Content. 401 Unauthorized – no valid token. 403 Forbidden – caller is not Admin or the event's Organiser. 404 Not Found – event does not exist.
GET	/api/events/{eventId}/enrolments	Retrieves all participants enrolled in an event.	Admin, Organiser	None	200 OK. Returns enrolment records with participant and category information. 401 Unauthorized – no valid token. 403 Forbidden – caller is not Admin or the event's Organiser. 404 Not Found – event does not exist.
GET	/api/events/{eventId}/categories	Retrieves all categories belonging to an event.	Public	None	200 OK. Returns the categories associated with the event. 404 Not Found – event does not exist.
GET	/api/categories/{categoryId}	Retrieves details for a specific category.	Public	None	200 OK. Returns CategoryID, EventID, Name, DistanceKm, MaxParticipants and EntryFee. 404 Not Found – category does not exist.
POST	/api/events/{eventId}/categories	Creates a category for an event.	Admin, Organiser	Name, DistanceKm, MaxParticipants, EntryFee	201 Created. Returns the new CategoryID and category details. 400 Bad Request – validation errors. 401 Unauthorized – no valid token. 403 Forbidden – caller is not Admin or the event's Organiser. 404 Not Found – event does not exist.

PUT	/api/categories/{categoryId}	Updates an existing event category.	Admin, Organiser	Name, DistanceKm, MaxParticipants, EntryFee	200 OK. Returns the updated category. 400 Bad Request – validation errors. 401 Unauthorized – no valid token. 403 Forbidden – caller is not Admin or the event's Organiser. 404 Not Found – category does not exist.
DELETE	/api/categories/{categoryId}	Removes a category from an event.	Admin, Organiser	None	204 No Content. 401 Unauthorized – no valid token. 403 Forbidden – caller is not Admin or the event's Organiser. 404 Not Found – category does not exist. 409 Conflict – category has existing enrolments.
POST	/api/events/{eventId}/enrolments	Enrols the authenticated participant into a category for an event.	Participant	CategoryID	201 Created. Returns EnrolmentID, UserID, CategoryID, EnrolmentDate, Status. 400 Bad Request – validation errors. 401 Unauthorized – no valid token. 404 Not Found – event or category does not exist. 409 Conflict – already enrolled in this category, or category is full.
GET	/api/enrolments	Retrieves the authenticated user's enrolments.	Participant	None	200 OK. Returns a list of the user's enrolments. 401 Unauthorized – no valid token.

GET	/api/enrolments/{enrolmentId}	Retrieves details about a specific enrolment.	Participant, Admin, Organiser	None	200 OK. Returns the enrolment details. 401 Unauthorized – no valid token. 403 Forbidden – Participant does not own this enrolment. 404 Not Found – enrolment does not exist.
PUT	/api/enrolments/{enrolmentId}	Updates the status of an enrolment.	Admin, Organiser	Status	200 OK. Returns the updated enrolment. 400 Bad Request – invalid status value. 401 Unauthorized – no valid token. 403 Forbidden – caller is not Admin or the event's Organiser. 404 Not Found – enrolment does not exist.
DELETE	/api/enrolments/{enrolmentId}	Cancels an enrolment.	Participant, Admin, Organiser	None	204 No Content. 401 Unauthorized – no valid token. 403 Forbidden – Participant does not own this enrolment. 404 Not Found – enrolment does not exist.
POST	/api/enrolments/{enrolmentId}/results	Records the result for a participant after an event.	Admin, Organiser	FinishTime, Position	201 Created. Returns ResultID, EnrolmentID, FinishTime, Position and RecordedAt. 400 Bad Request – validation errors. 401 Unauthorized – no valid token. 403 Forbidden – caller is not Admin or the event's Organiser. 404 Not Found – enrolment does not exist. 409 Conflict – result already recorded for this enrolment.

GET	/api/results/{resultId}	Retrieves a specific result.	Participant (own result), Admin, Organiser	None	200 OK. Returns result information and the associated enrolment. 401 Unauthorized – no valid token. 403 Forbidden – Participant does not own this result. 404 Not Found – result does not exist.
PUT	/api/results/{resultId}	Updates a participant's result.	Admin, Organiser	FinishTime, Position	200 OK. Returns the updated result. 400 Bad Request – validation errors. 401 Unauthorized – no valid token. 403 Forbidden – caller is not Admin or the event's Organiser. 404 Not Found – result does not exist.
DELETE	/api/results/{resultId}	Removes a recorded result.	Admin	None	204 No Content. 401 Unauthorized – no valid token. 403 Forbidden – caller is not an Admin. 404 Not Found – result does not exist.
GET	/api/enrolments/{enrolmentId}/results	Retrieves the result associated with a specific enrolment.	Participant, Admin, Organiser	None	200 OK. Returns ResultID, FinishTime, Position and RecordedAt. 401 Unauthorized – no valid token. 403 Forbidden – Participant does not own this enrolment. 404 Not Found – enrolment or result does not exist.
GET	/api/events/{eventId}/results	Retrieves all results for an event.	Admin, Organiser	None	200 OK. Returns a ranked list of participants and their results. 401 Unauthorized – no valid token. 403 Forbidden – caller is not Admin or the event's Organiser. 404 Not Found – event does not exist.

GET	/api/users/me/results	Retrieves all results belonging to the authenticated participant.	Participant	None	200 OK. Returns the participant's results across events. 401 Unauthorized – no valid token.
GET	/api/venues	Retrieves all venues.	Public	None	200 OK. Returns venue details and location information.
GET	/api/venues/{venueId}	Retrieves a specific venue.	Public	None	200 OK. Returns venue name, address, city, province, latitude and longitude. 404 Not Found – venue does not exist.
POST	/api/venues	Creates a new venue.	Admin	Name, AddressLine, City, Province, Latitude, Longitude	201 Created. Returns the new VenueID and venue details. 400 Bad Request – validation errors. 401 Unauthorized – no valid token. 403 Forbidden – caller is not an Admin.
PUT	/api/venues/{venueId}	Updates venue information.	Admin	Venue fields to update	200 OK. Returns the updated venue. 400 Bad Request – validation errors. 401 Unauthorized – no valid token. 403 Forbidden – caller is not an Admin. 404 Not Found – venue does not exist.
DELETE	/api/venues/{venueId}	Removes a venue.	Admin	None	204 No Content. 401 Unauthorized – no valid token. 403 Forbidden – caller is not an Admin. 404 Not Found – venue does not exist. 409 Conflict – venue has associated events.